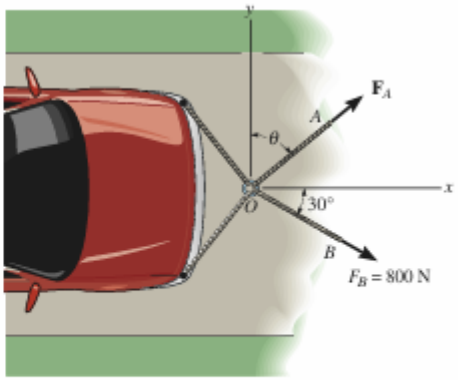
Assignment (2) Model Answer

1.

Determine the magnitude and direction, measured counterclockwise from the positive x axis, of the resultant force acting on the ring at O, if $F_A = 750$ N and $\theta = 45°$.



SOLUTION

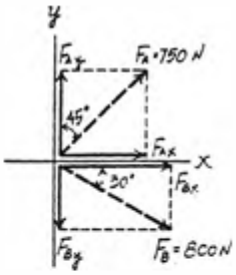
Scalar Notation: Suming the force components algebraically, we have

$$\stackrel{+}{\rightarrow} F_{R_x} = \Sigma F_x; \qquad F_{R_x} = 750 \sin 45° + 800 \cos 30°$$

$$= 1223.15 \text{ N} \rightarrow$$

$$+\uparrow F_{R_y} = \Sigma F_y; \qquad F_{R_y} = 750 \cos 45° - 800 \sin 30°$$

$$= 130.33 \text{ N} \uparrow$$



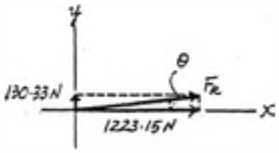
The magnitude of the resultant force $\mathbf{F}_R$ is

$$F_R = \sqrt{F_{R_x}^2 + F_{R_y}^2}$$

$$= \sqrt{1223.15^2 + 130.33^2} = 1230 \text{ N} = 1.23 \text{ kN} \qquad \textbf{Ans.}$$

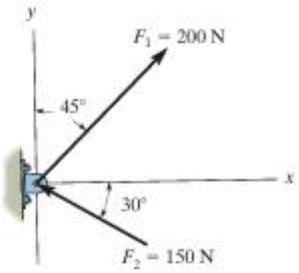
The directional angle θ measured counterclockwise from positive x axis is

$$\theta = \tan^{-1}\frac{F_{R_y}}{F_{R_x}} = \tan^{-1}\left(\frac{130.33}{1223.15}\right) = 6.08° \qquad \textbf{Ans.}$$



2.

Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive x axis.



SOLUTION

Scalar Notation. Summing the force components along x and y axes algebraically by referring to Fig. a,

$$\stackrel{+}{\rightarrow}(F_R)_x = \Sigma F_x; \qquad (F_R)_x = 200\sin 45° - 150\cos 30° = 11.518\text{ N} \rightarrow$$

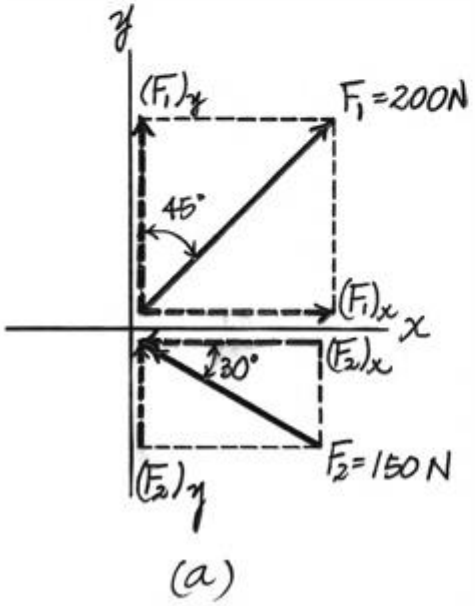
$$+\uparrow(F_R)_y = \Sigma F_y; \qquad (F_R)_y = 200\cos 45° + 150\sin 30° = 216.42\text{ N} \uparrow$$

Referring to Fig. b, the magnitude of the resultant force F_R is

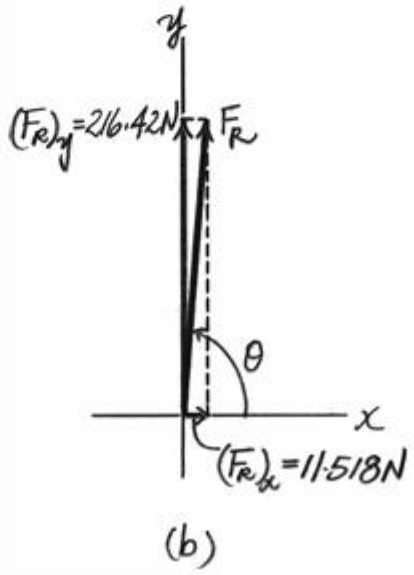
$$F_R = \sqrt{(F_R)_x^2 + (F_R)_y^2} = \sqrt{11.518^2 + 216.42^2} = 216.73\text{ N} = 217\text{ N} \qquad \textbf{Ans.}$$

And the directional angle θ of $\mathbf{F}_R$ measured counterclockwise from the positive x axis is

$$\theta = \tan^{-1}\left[\frac{(F_R)_y}{(F_R)_x}\right] = \tan^{-1}\left(\frac{216.42}{11.518}\right) = 86.95° = 87.0° \qquad \textbf{Ans.}$$



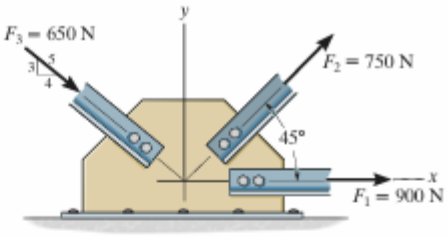
(a)



(b)

3.

Determine the magnitude of the resultant force acting on the plate and its direction, measured counterclockwise from the positive x axis.



SOLUTION

Rectangular Components: By referring to Fig. *a*, the x and y components of $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ can be written as

$$(F_1)_x = 900\ \text{N} \qquad (F_1)_y = 0$$

$$(F_2)_x = 750\cos 45^\circ = 530.33\ \text{N} \qquad (F_2)_y = 750\sin 45^\circ = 530.33\ \text{N}$$

$$(F_3)_x = 650\left(\frac{4}{5}\right) = 520\ \text{N} \qquad (F_3)_y = 650\left(\frac{3}{5}\right) = 390\ \text{N}$$

Resultant Force: Summing the force components algebraically along the x and y axes, we have

$$\stackrel{+}{\rightarrow} \Sigma(F_R)_x = \Sigma F_x; \qquad (F_R)_x = 900 + 530.33 + 520 = 1950.33\ \text{N} \rightarrow$$

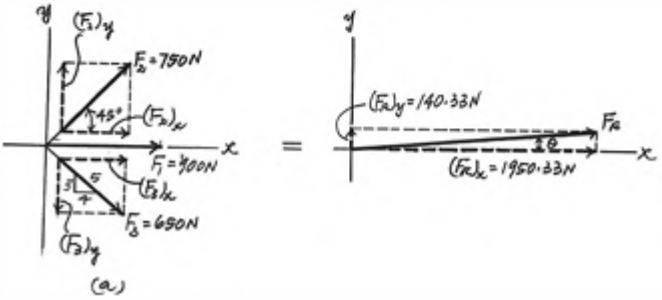
$$+\uparrow \Sigma(F_R)_y = \Sigma F_y; \qquad (F_R)_y = 530.33 - 390 = 140.33\ \text{N} \uparrow$$

The magnitude of the resultant force $\mathbf{F}_R$ is

$$F_R = \sqrt{(F_R)_x^2 + (F_R)_y^2} = \sqrt{1950.33^2 + 140.33^2} = 1955\ \text{N} = 1.96\ \text{kN}\quad \textbf{Ans.}$$

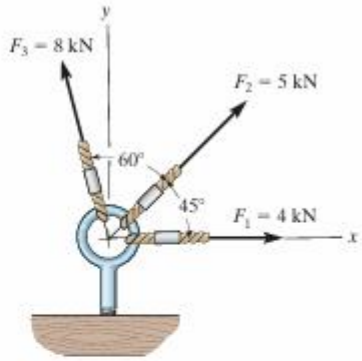
The direction angle θ of $\mathbf{F}_R$, measured clockwise from the positive x axis, is

$$\theta = \tan^{-1}\left[\frac{(F_R)_y}{(F_R)_x}\right] = \tan^{-1}\left(\frac{140.33}{1950.33}\right) = 4.12^\circ \qquad \textbf{Ans.}$$



4.

Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive x axis.



SOLUTION

Scalar Notation. Summing the force components along x and y axes algebraically by referring to Fig. a,

$$\stackrel{+}{\rightarrow}(F_R)_x = \Sigma F_x; \qquad (F_R)_x = 4 + 5\cos 45° - 8\sin 15° = 5.465 \text{ kN} \rightarrow$$

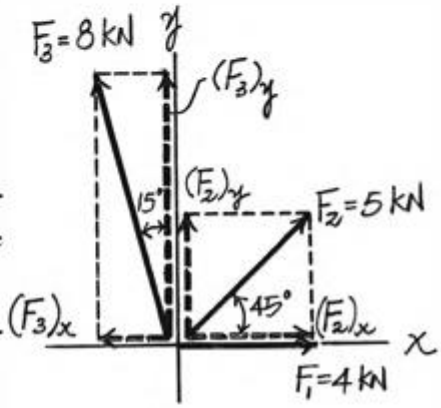
$$+\uparrow(F_R)_y = \Sigma F_y; \qquad (F_R)_y = 5\sin 45° + 8\cos 15° = 11.263 \text{ kN} \uparrow$$

By referring to Fig. b, the magnitude of the resultant force $\mathbf{F}_R$ is

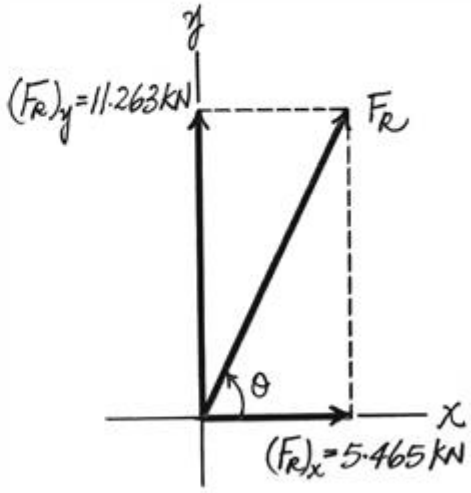
$$F_R = \sqrt{(F_R)_x^2 + (F_R)_y^2} = \sqrt{5.465^2 + 11.263^2} = 12.52 \text{ kN} = 12.5 \text{ kN} \quad \textbf{Ans.}$$

And the directional angle θ of $\mathbf{F}_R$ measured counterclockwise from the positive x axis is

$$\theta = \tan^{-1}\left[\frac{(F_R)_y}{(F_R)_x}\right] = \tan^{-1}\left(\frac{11.263}{5.465}\right) = 64.12° = 64.1° \quad \textbf{Ans.}$$



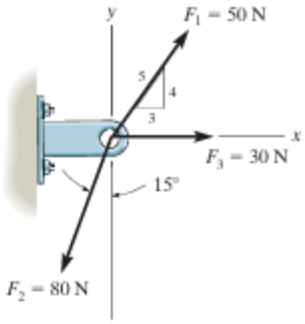
(a)



(b)

5.

Express each of the three forces acting on the support in Cartesian vector form and determine the magnitude of the resultant force and its direction, measured clockwise from positive x axis.



SOLUTION

Cartesian Notation. Referring to Fig. *a*,

$$\mathbf{F}_1 = (F_1)_x\,\mathbf{i} + (F_1)_y\,\mathbf{j} = 50\left(\frac{3}{5}\right)\mathbf{i} + 50\left(\frac{4}{5}\right)\mathbf{j} = \{30\,\mathbf{i} + 40\,\mathbf{j}\}\ \text{N} \qquad \textbf{Ans.}$$

$$\mathbf{F}_2 = -(F_2)_x\,\mathbf{i} - (F_2)_y\,\mathbf{j} = -80\sin 15°\,\mathbf{i} - 80\cos 15°\,\mathbf{j}$$

$$= \{-20.71\,\mathbf{i} - 77.27\,\mathbf{j}\}\ \text{N}$$

$$= \{-20.7\,\mathbf{i} - 77.3\,\mathbf{j}\}\ \text{N} \qquad \textbf{Ans.}$$

$$F_3 = (F_3)_x\,\mathbf{i} = \{30\,\mathbf{i}\} \qquad \textbf{Ans.}$$

Thus, the resultant force is

$$\mathbf{F}_R = \Sigma\mathbf{F}; \qquad \mathbf{F}_R = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$$

$$= (30\mathbf{i} + 40\,\mathbf{j}) + (-20.71\mathbf{i} - 77.27\mathbf{j}) + 30\mathbf{i}$$

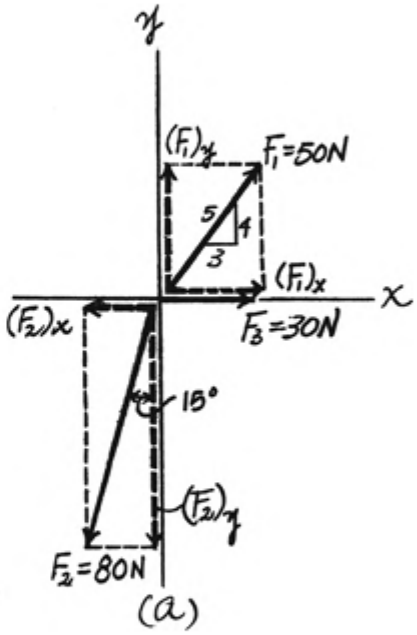
$$= \{39.29\,\mathbf{i} - 37.27\,\mathbf{j}\}\ \text{N}$$

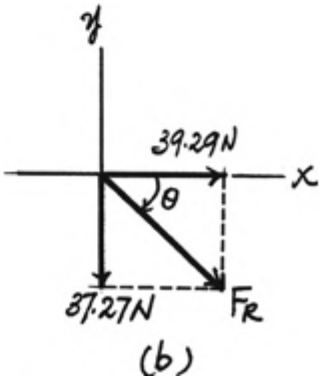
Referring to Fig. *b*, the magnitude of $\mathbf{F}_R$ is

$$F_R = \sqrt{39.29^2 + 37.27^2} = 54.16\ \text{N} = 54.2\ \text{N} \qquad \textbf{Ans.}$$

And its directional angle θ measured clockwise from the positive x axis is

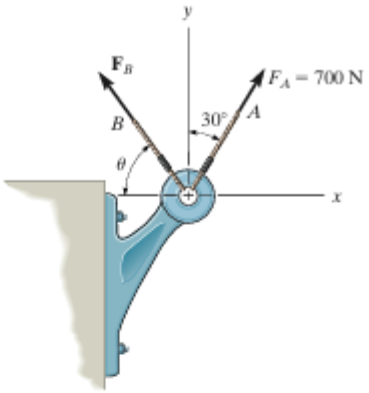
$$\theta = \tan^{-1}\left(\frac{37.27}{39.29}\right) = 43.49° - 43.5° \qquad \textbf{Ans.}$$





6.

Determine the magnitude and orientation θ of $\mathbf{F}_B$ so that the resultant force is directed along the positive y axis and has a magnitude of 1500 N.



SOLUTION

Scalar Notation: Suming the force components algebraically, we have

$\overset{+}{\rightarrow} F_{R_x} = \Sigma F_x;$ $\quad 0 = 700 \sin 30° - F_B \cos\theta$

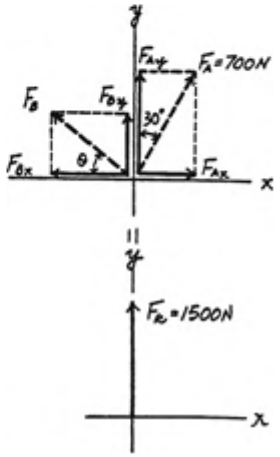
$$F_B \cos\theta = 350 \qquad \textbf{(1)}$$

$+\uparrow F_{R_y} = \Sigma F_y;$ $\quad 1500 = 700 \cos 30° + F_B \sin\theta$

$$F_B \sin\theta = 893.8 \qquad \textbf{(2)}$$

Solving Eq. (1) and (2) yields

$$\theta = 68.6° \qquad F_B = 960 \text{ N} \qquad \textbf{Ans.}$$



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